from zero to hero

AI Notes

by hand, in 8 pages

☑ what AI is ☑ how it learns
☑ chatbots ☑ prompts ☑ agents
☑ vibe coding

save this one →

@evelineclarince

1. What is AI?

Computers doing tasks we link to intelligence

Circles inside circles

AI → Machine learning → Deep learning → Generative AI

AI

- Tasks we link to human intelligence
- Search, maps, spam filters, voice assistants
- Chatbots are just one kind

Machine learning

- Programs improve at a task from data
- You give examples, not rules

Deep learning

- Many layers of neurons
- Low layers: edges
- High layers: faces, digits

Generative AI

- Creates and edits media
- Text, images, sound
- Drove the 2020s AI boom

Remember

AI finds patterns in data. Patterns, not magic. So it can be wrong.

2. How machines learn

Three ways

Type	Data	Example
Supervised	Labeled with answers	Spam or not spam
Unsupervised	No labels	Grouping similar things
Reinforcement	Rewards, penalties	Playing Go (AlphaGo)

The learning loop

Guess → Compare → Adjust → Repeat

Training

- Show lots of examples
- Measure the error
- Nudge the numbers a tiny bit

Testing

- Use examples it never saw
- Right on new ones? It learned
- Only on old ones? It memorized

Remember

Bad data in, bad answers out.

3. Neural networks

No math, promise

The shape

Input → Hidden layer → Hidden layer → Output

The parts

- Input: numbers go in
- Hidden: at least one layer
- Output: a guess comes out
- Deep = 2+ hidden layers

Weights

- Numbers on the connections
- They decide what passes on
- Learning = setting the weights

How it trains

Make a guess → Measure the error → Backpropagation → Weights change

Example: reading an image

- An image is just numbers
- Early layers: edges and blobs of color
- Later layers: bigger areas and concepts

4. How chatbots work

LLMs, tokens, next word

The core idea



Word	Meaning
Token	A small piece of text
Context window	Its working memory
Pretraining	Learn to predict the next word
Fine-tuning	Adjust for a specific job
RLHF	Humans rank answers, model improves
Temperature	How random the answers are
RAG	Fetch documents, then answer from them

Remember

It predicts likely text. Likely is not the same as true.

5. Prompts that work

Treat AI like a brilliant new hire

The template

- Role: who should it be?
- Context: what's going on, and why?
- Task: what exactly do you want?
- Examples: 3 to 5 good ones
- Format: how should it look?

Do

- Explain the why
- Say what to do, not only what to avoid
- Long documents first, question last
- Draft, review, refine

Avoid

- Vague asks
- No context
- Many tasks in one prompt
- Trusting without checking

Colleague test

Show your prompt to a friend. If they'd be confused, so will the AI.

6. When AI makes things up

Hallucinations

What is it?

False or misleading information presented as fact.

Why?

- Trained to guess the next word
- Rewarded for guessing, not doubt
- Flawed training data
- Tuned to be agreeable

Real case, 2023

- Lawyers cited 6 fake cases
- ChatGPT had invented them
- Case dismissed
- $5,000 fine

Protect yourself

Ask for sources → Open each one → Check numbers → Allow I don't know

Remember

Confident is not correct. You sign it, you own it.

7. AI agents + MCP

From answers to actions

How an agent works

Goal → Plan → Use a tool → Check result → Finish or repeat

Parts

Part	Job
Model	Decides next step
Tools	Web, files, code
Memory	Keeps context
Limits	Keep it safe

What is MCP?

- Model Context Protocol
- Open standard from Anthropic
- One way to plug tools and data into AI apps

Stay safe

- Give only the access it needs
- A human approves big actions
- Pages it reads can hide instructions

8. Vibe coding

Building by describing

The loop

Describe → AI writes code → Run it → Refine → Ship

Great for

- Prototypes
- Personal tools
- Learning
- Small websites

Careful with

- Logins and payments
- Private data
- Code nobody read
- Secrets in the code

Before you ship

- Plan first, build one step at a time
- Give it a test it can run
- Use Git, so you can undo
- Get a review from fresh eyes

Where it came from

Term coined by Andrej Karpathy in February 2025.